

Lie Detection via Micro Expression

2025.06.06

ISOO

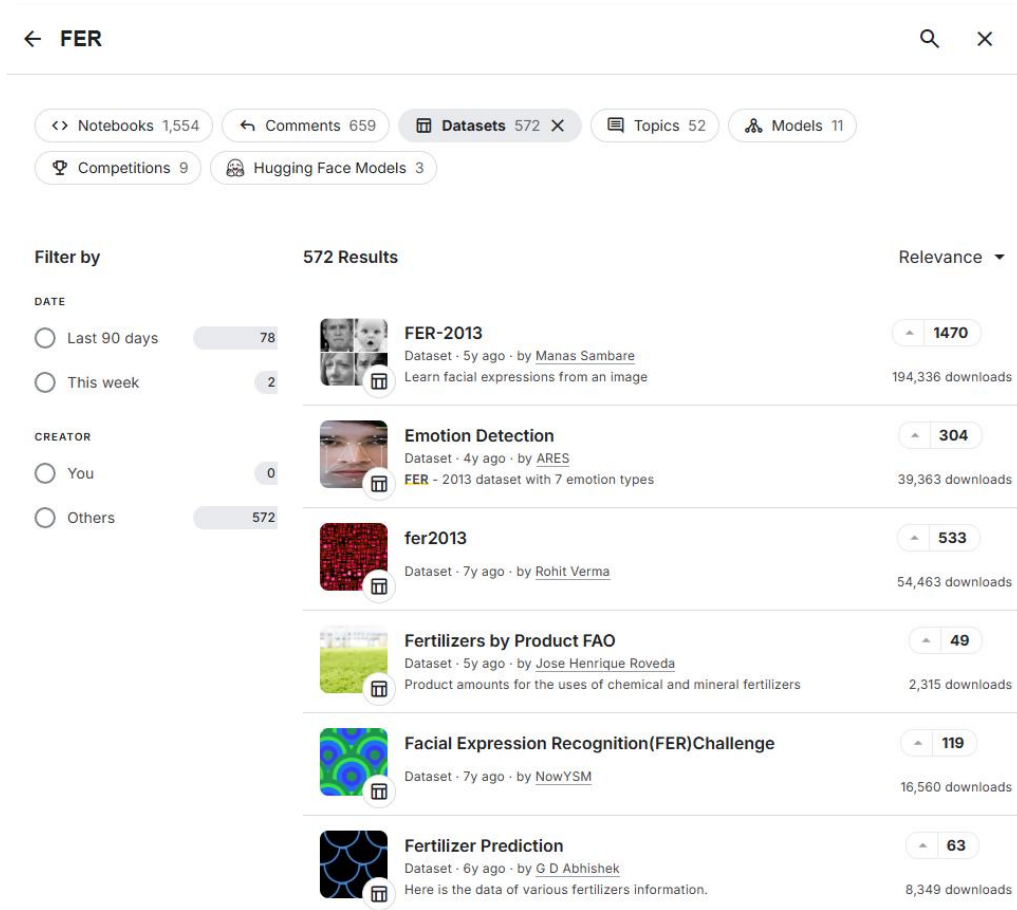
김준하, 노윤현, 윤지현, 조해나, 최우석

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-

1. Motivation & Goal

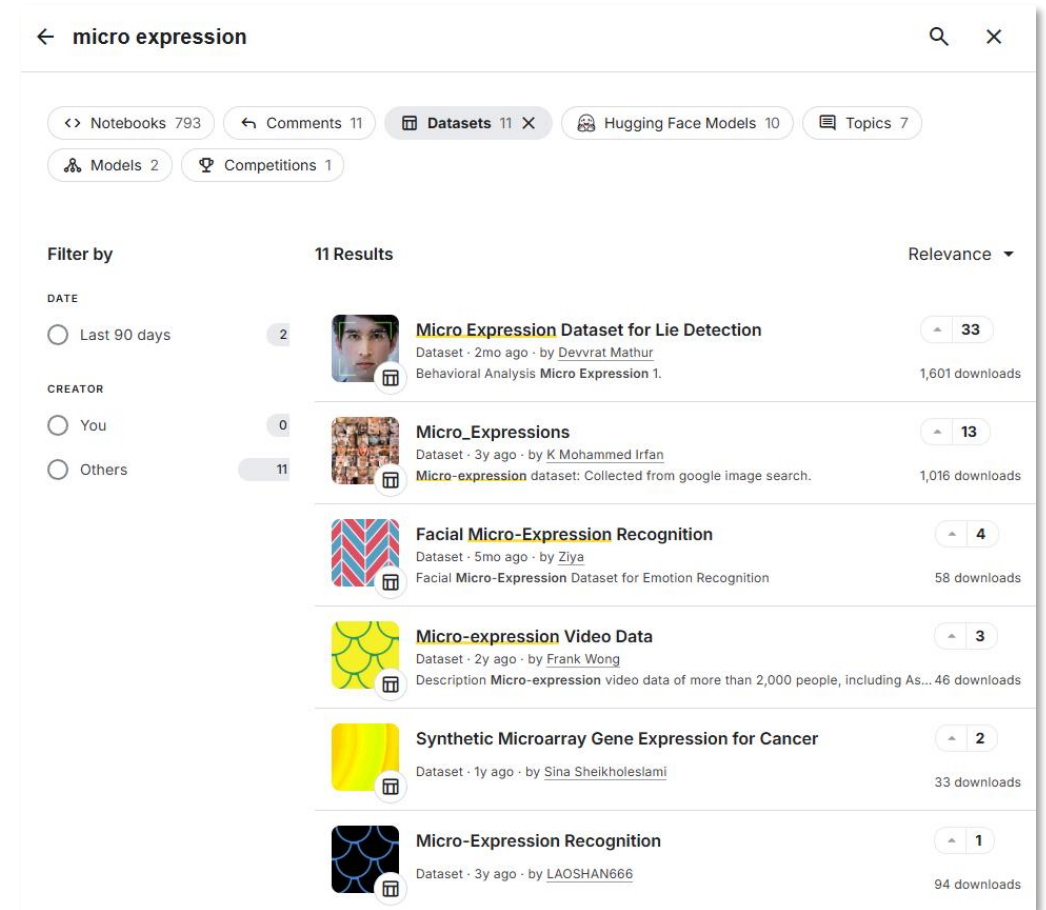
1. Motivation & Goal



The screenshot shows the Kaggle search results for 'FER' (Facial Expression Recognition) datasets. The page has a header with a back arrow, the search term 'FER', and search and close icons. Below the header, there are filters for Notebooks (1,554), Comments (659), Datasets (572), Topics (52), Models (11), Competitions (9), and Hugging Face Models (3). The 'Filter by' section on the left shows 'DATE' with options for 'Last 90 days' (78 results) and 'This week' (2 results), and 'CREATOR' with options for 'You' (0 results) and 'Others' (572 results). The main content area displays a list of 572 results, with the first six visible:

Dataset Icon	Dataset Name	Dataset Description	Relevance	Downloads
	FER-2013	Dataset · 5y ago · by Manas Sambare Learn facial expressions from an image	1470	194,336 downloads
	Emotion Detection	Dataset · 4y ago · by ARES FER - 2013 dataset with 7 emotion types	304	39,363 downloads
	fer2013	Dataset · 7y ago · by Rohit Verma	533	54,463 downloads
	Fertilizers by Product FAO	Dataset · 5y ago · by Jose Henrique Roveda Product amounts for the uses of chemical and mineral fertilizers	49	2,315 downloads
	Facial Expression Recognition(FER)Challenge	Dataset · 7y ago · by NowYSM	119	16,560 downloads
	Fertilizer Prediction	Dataset · 6y ago · by G D Abhishek Here is the data of various fertilizers information.	63	8,349 downloads

FER datasets in kaggle

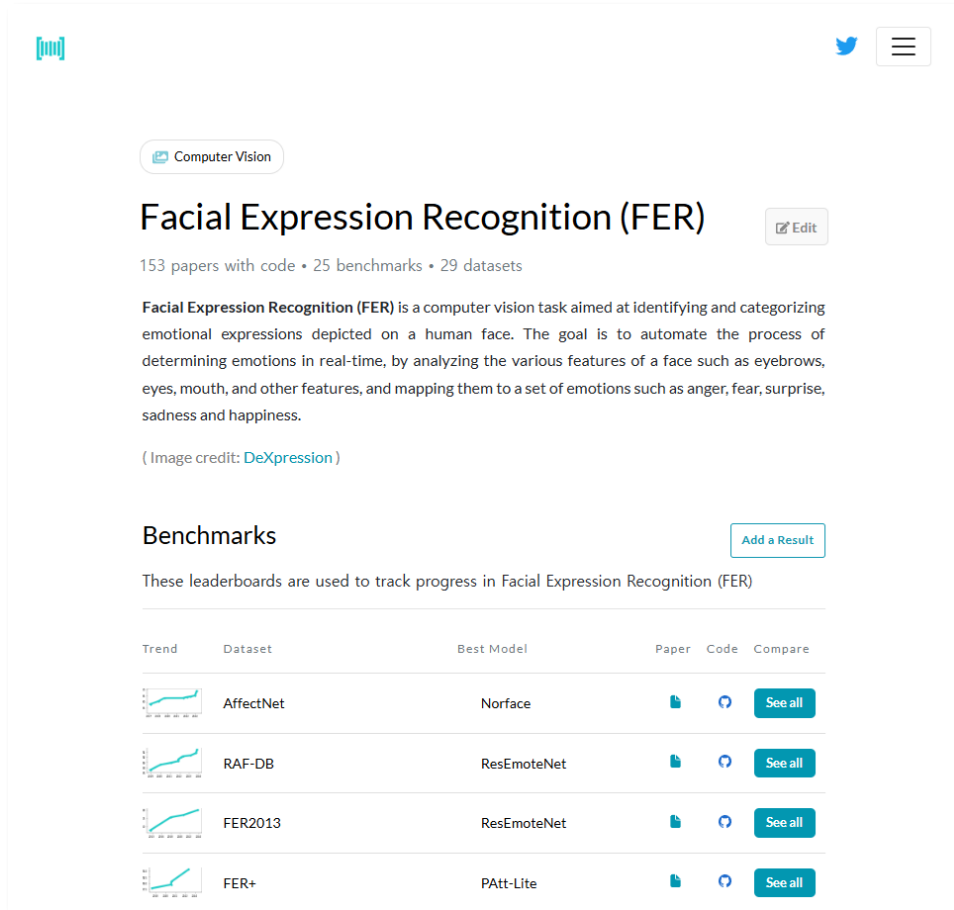


The screenshot shows the Kaggle search results for 'micro expression' datasets. The page has a header with a back arrow, the search term 'micro expression', and search and close icons. Below the header, there are filters for Notebooks (793), Comments (11), Datasets (11), Hugging Face Models (10), Topics (7), Models (2), and Competitions (1). The 'Filter by' section on the left shows 'DATE' with options for 'Last 90 days' (2 results) and 'Others' (11 results), and 'CREATOR' with options for 'You' (0 results) and 'Others' (11 results). The main content area displays a list of 11 results, with the first six visible:

Dataset Icon	Dataset Name	Dataset Description	Relevance	Downloads
	Micro Expression Dataset for Lie Detection	Dataset · 2mo ago · by Devvrat Mathur Behavioral Analysis Micro Expression 1.	33	1,601 downloads
	Micro_Expressions	Dataset · 3y ago · by K Mohammed Irfan Micro-expression dataset: Collected from google image search.	13	1,016 downloads
	Facial Micro-Expression Recognition	Dataset · 5mo ago · by Ziya Facial Micro-Expression Dataset for Emotion Recognition	4	58 downloads
	Micro-expression Video Data	Dataset · 2y ago · by Frank Wong Description Micro-expression video data of more than 2,000 people, including As...	3	46 downloads
	Synthetic Microarray Gene Expression for Cancer	Dataset · 1y ago · by Sina Sheikholeslami	2	33 downloads
	Micro-Expression Recognition	Dataset · 3y ago · by LAOSHAN666	1	94 downloads

Micro-expression datasets in kaggle

1. Motivation & Goal



Computer Vision

Facial Expression Recognition (FER)

153 papers with code • 25 benchmarks • 29 datasets

Facial Expression Recognition (FER) is a computer vision task aimed at identifying and categorizing emotional expressions depicted on a human face. The goal is to automate the process of determining emotions in real-time, by analyzing the various features of a face such as eyebrows, eyes, mouth, and other features, and mapping them to a set of emotions such as anger, fear, surprise, sadness and happiness.

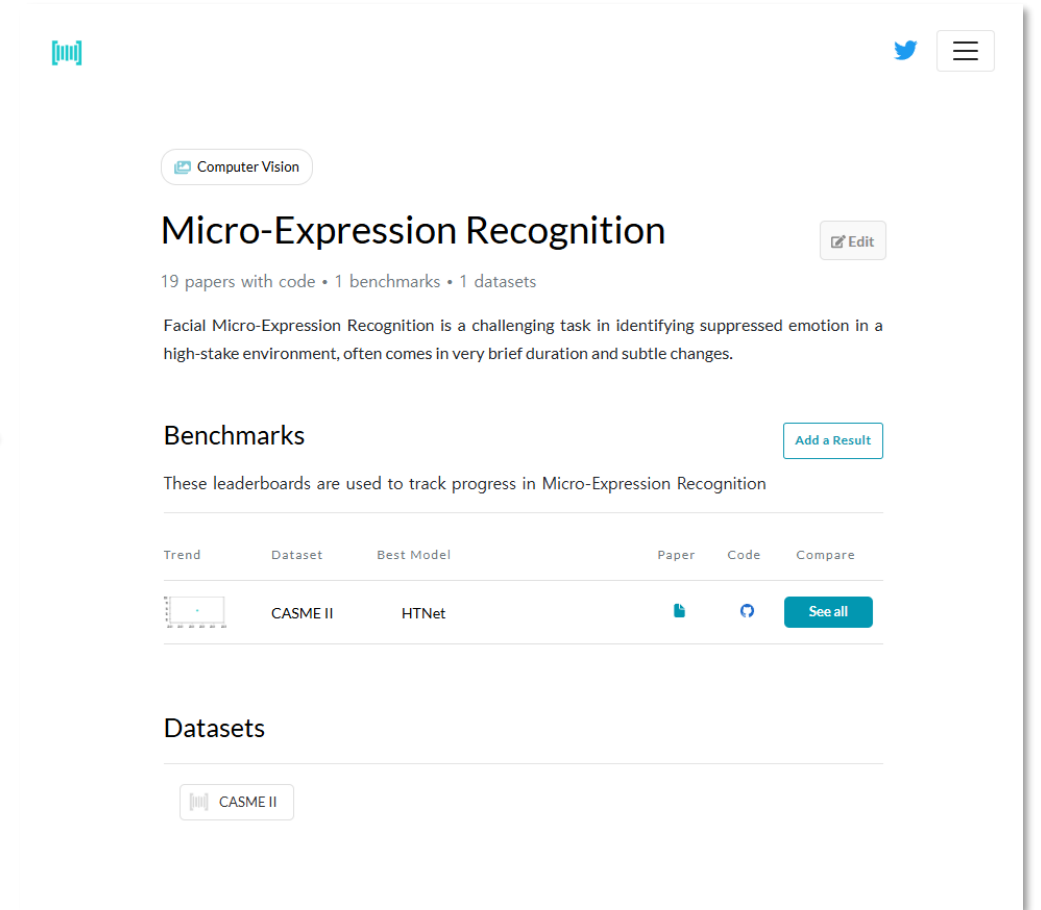
(Image credit: DeXpression)

Benchmarks

These leaderboards are used to track progress in Facial Expression Recognition (FER)

Trend	Dataset	Best Model	Paper	Code	Compare
	AffectNet	Norface			See all
	RAF-DB	ResEmoteNet			See all
	FER2013	ResEmoteNet			See all
	FER+	PAtt-Lite			See all

FER researches in paperswithcode



Computer Vision

Micro-Expression Recognition

19 papers with code • 1 benchmarks • 1 datasets

Facial Micro-Expression Recognition is a challenging task in identifying suppressed emotion in a high-stake environment, often comes in very brief duration and subtle changes.

Benchmarks

These leaderboards are used to track progress in Micro-Expression Recognition

Trend	Dataset	Best Model	Paper	Code	Compare
	CASME II	HTNet			See all

Datasets

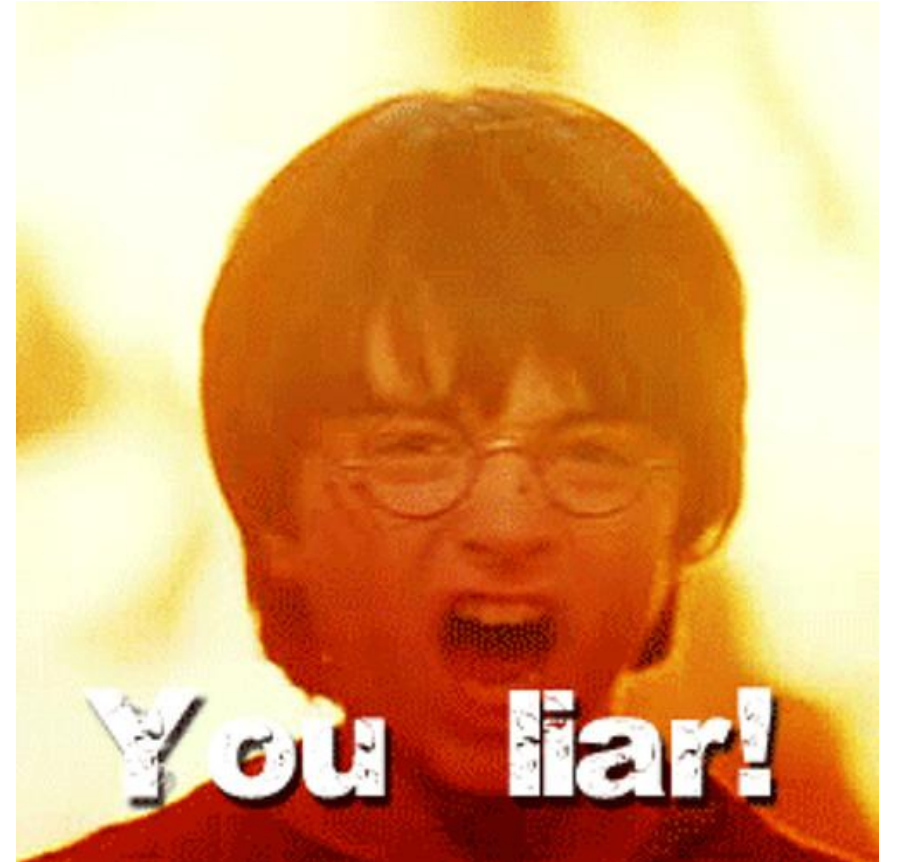
CASME II

Micro-Expression researches in paperswithcode

1. Motivation & Goal

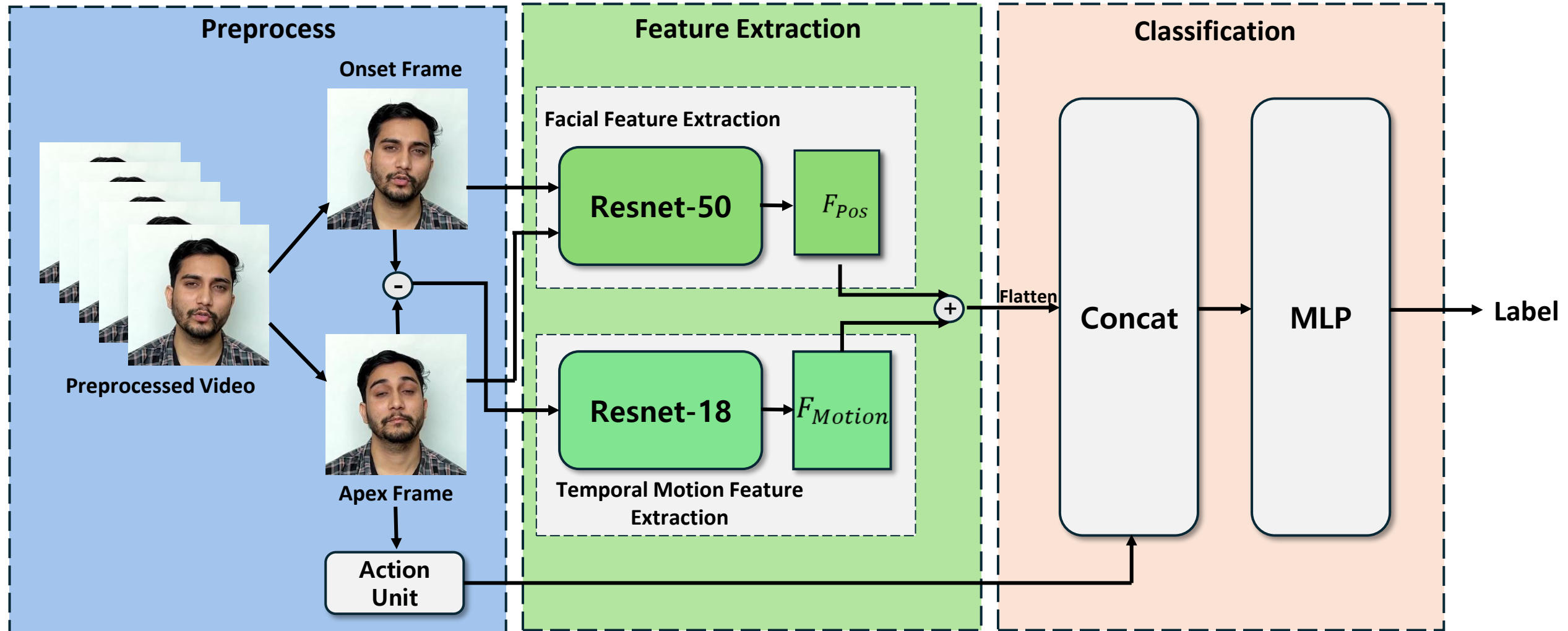
Goal

- Detect a liar by using micro expression



2. Previous Progress

2. Previous Progress



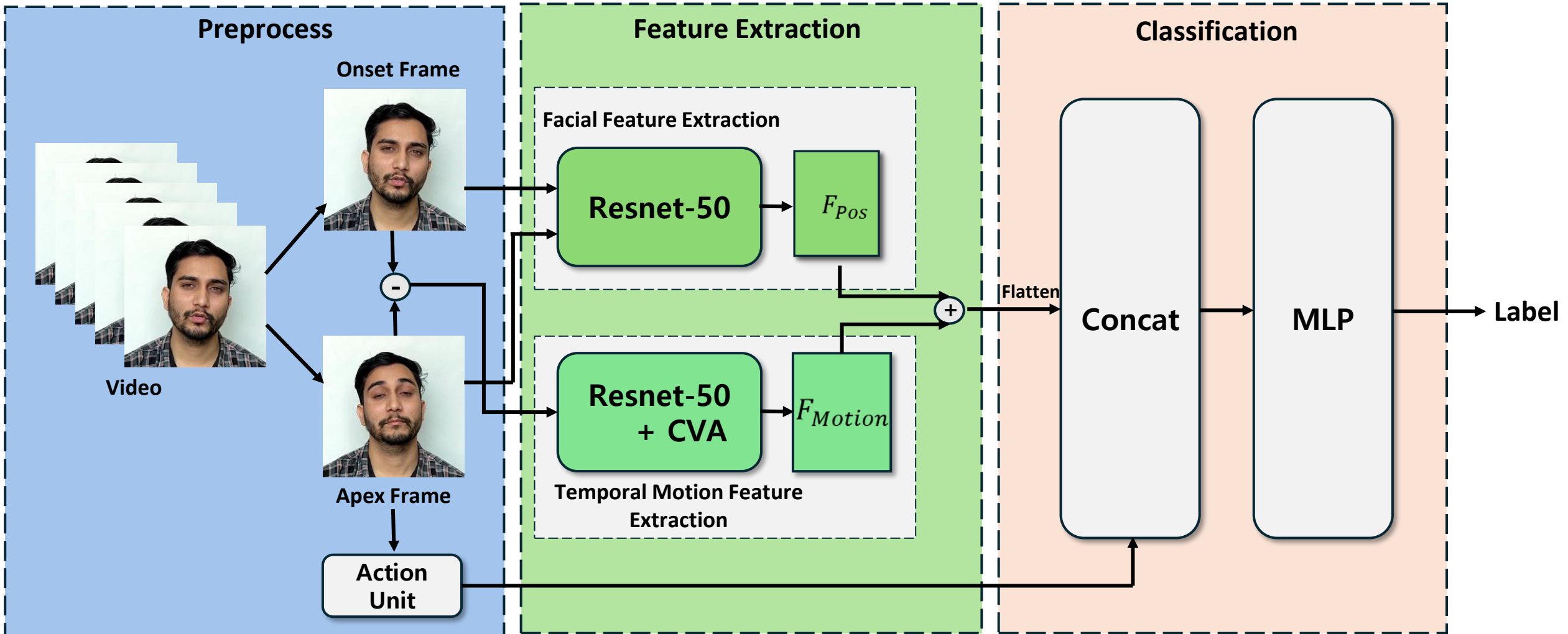
2. Previous Progress

Previous Plans

- Step: Preprocessing → Feature Extraction → Binary Classification → Evaluation
 - Preprocessing: Use additional datasets to make final datasets.
 - Feature Extraction : Test other methods and choose final methods.
 - Binary Classification: Test other methods and choose final methods and fine tuning.
 - Evaluation: Choose classification evaluation method.
-

3. Progress After Interim

3. Progress After Interim



3. Progress After Interim

Data Preparation & Preprocessing

- Kaggle의 거짓말 탐지를 위한 Micro Expression 데이터셋 활용
- Real-life deception (법정 위증 데이터셋) 활용
- Bag-of-Lies 데이터셋 활용하려 했지만 실패
- 총 170개의 train data 사용

Micro Expression Dataset for Lie Detection

Detailed Facial Micro-expression Image Dataset for Lie Detection

Real-life **Deception**

June 15, 2016

A multimodal dataset consisting of real-life **deception**: deceptive and truthful trial testimonies, manually transcribed and annotated. The dataset includes 121 short videos, along with their transcriptions and gesture annotations. ([download](#))

Deception Detection Using Real-life Trial Data ([BibTeX](#))

Verónica Pérez-Rosas, Mohamed Abouelenen, Rada Mihalcea, Mihai Burzo

17th ACM International Conference on Multimodal Interaction (ICMI), 2015

3. Progress After Interim

Onset/Apex Frame Extraction

- mediapipe 라이브러리 중 Face Mesh를 이용해 얼굴 랜드마크 추출
 - 이전 이미지 프레임과 현재 이미지 프레임 간의 픽셀 움직임 변화량 (motion score) 계산
 - Optical flow와 landmark displacement에 0.7:0.3의 가중치 부여
- ```
Combined Score
combined_score = mag_flow * 0.7 + landmark_motion * 0.3
return combined_score
```
- threshold 이상의 변화량이 나타나는 프레임을 onset(없을 시 첫번째 프레임)으로 지정
  - motion score가 최댓값이 되는 프레임을 apex로 지정



Onset Frame



Apex Frame

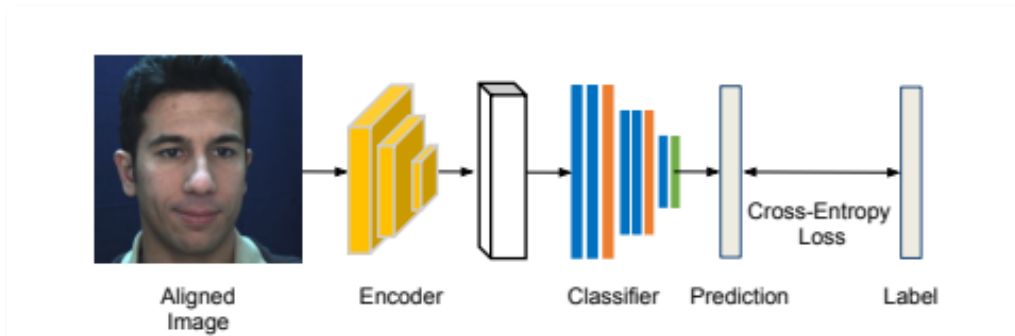
# 3. Progress After Interim

## AU Extraction - LibreFace

- LibreFace: An Open-Source Toolkit for Deep Facial Expression Analysis
  - 얼굴 인식 → 정렬 → CNN 기반 딥러닝 모델에 적용
  - LibreFace가 인식할 수 있는 17개의 AU Unit에 대해 presence(binary)와 intensity(continuous) 예측 (Table 2.) → 총 29-dimension

Table 2. Available action units in LibreFace. **R** - LibreFace provides intensity estimation, **D** - LibreFace provides detection of occurrence.

| AU   | FACS name            | Prediction |
|------|----------------------|------------|
| AU1  | Inner brow raiser    | R          |
| AU2  | Outer brow raiser    | R          |
| AU4  | Brow lowerer         | R          |
| AU5  | Upper lid raiser     | R          |
| AU6  | Cheek raiser         | R          |
| AU7  | Lid tightener        | D          |
| AU9  | Nose wrinkler        | R          |
| AU10 | Upper lip raiser     | D          |
| AU12 | Lip corner puller    | R          |
| AU14 | Dimpler              | D          |
| AU15 | Lip corner depressor | R          |
| AU17 | Chin raiser          | R          |
| AU20 | Lip stretcher        | R          |
| AU23 | Lip tightener        | D          |
| AU24 | Lip pressor          | D          |
| AU25 | Lips part            | R          |
| AU26 | Jaw drop             | R          |



|   | A          | B          | C          | D          | E          | F          | G          | H          | I               | J              |
|---|------------|------------|------------|------------|------------|------------|------------|------------|-----------------|----------------|
| 1 | au_1       | au_2       | au_4       | au_5       | au_6       | au_7       | au_9       | au_10      | au_12           | au_14          |
| 2 | 0          | 0          | 0          | 0          | 0          | 0          | 0          | 1          | 0               | 0              |
| 3 | au_15      | au_17      | au_20      | au_23      | au_24      | au_25      | au_26      | au_1_inten | au_2_inten      | au_4_intensity |
| 4 | 0          | 0          | 0          | 0          | 0          | 0          | 0          | 0.107      | 0.055           | 0.563          |
| 5 | au_5_inten | au_6_inten | au_9_inten | au_12_inte | au_15_inte | au_17_inte | au_20_inte | au_25_inte | au_26_intensity |                |
| 6 | 0.071      | 0.046      | 0.082      | 0.417      | 0.071      | 0.071      | 0.067      | 2.188      | 2.136           |                |

# 3. Progress After Interim

## AU Extraction - LibreFace

- 기존의 OpenFace 2.0(a facial behavior analysis toolkit) 툴보다 빠르고 정확하게 AU 판별 가능(Table 7.)
- 딥러닝 기반으로 복잡한 얼굴 움직임(micro-expression) 학습 가능
- GPU를 사용해 대량의 apex frame 이미지 처리에 적합 (Table 9.)
- 미세 표정을 기반으로 liar detection에 정량적인 AU feature 제공

Table 7. Efficiency Comparison between LibreFace and OpenFace 2.0 [4]. We report the average (Avg.) and standard deviation (Std.) of durations of each round of tests and averaged frame rate per second (FPS).

| Method                     | Avg.↓        | Std. | FPS↑         |
|----------------------------|--------------|------|--------------|
| OpenFace 2.0 [4] (AU only) | 50.11        | 0.22 | 19.96        |
| <b>LibreFace (AU only)</b> | <b>25.11</b> | 0.60 | <b>39.82</b> |
| <b>LibreFace (Ours)</b>    | 37.20        | 0.66 | 26.88        |

Table 9. Efficiency comparison among CPU-only mode and GPU mode in our proposed pipeline. We report the average (Avg.) and standard deviation (Std.) of durations of each round of test, and averaged frame rate per second (FPS)

| Method     | Avg.↓       | Std. | FPS↑          |
|------------|-------------|------|---------------|
| CPU-only   | 37.20       | 0.66 | 26.88         |
| <b>GPU</b> | <b>6.07</b> | 0.13 | <b>164.82</b> |

# 3. Progress After Interim

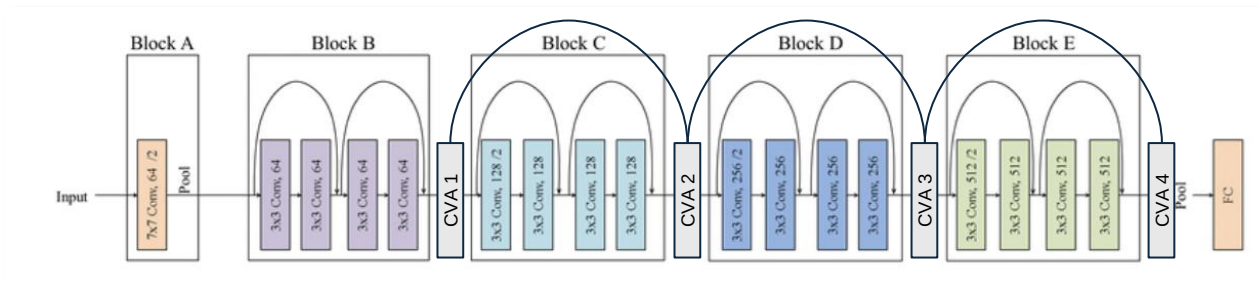
## Feature Extraction - TMFE

- CNN Model: ResNet 18 Model + CVA Module
  - Vertical한 특징을 강조하여 뽑기 위해 각 layer 별 CVA Module 추가
- Onset, Apex Frame을 element wise subtraction하여 input으로 사용

Onset Frame



Apex Frame



```
tensor([[[[2.3398e-03, 2.2910e-03, 1.9515e-03, ..., 1.5344e-03,
 1.6262e-03, 2.0991e-04],
 [0.0000e+00, 1.4738e-02, 0.0000e+00, ..., 0.0000e+00,
 2.5973e-03, 1.2227e-02],
 [5.1342e-03, 3.7192e-02, 2.3743e-02, ..., 2.0129e-02,
 1.1895e-02, 3.6625e-02],
```

```
...
[-2.5160e-02, -4.6538e-03, -3.8942e-03, ..., -2.4790e-03,
 -1.2922e-03, -6.4192e-02],
[-2.4224e-02, 0.0000e+00, -2.2408e-03, ..., -4.1906e-03,
 -3.0403e-03, -6.8799e-02],
[-4.4824e-03, -3.7549e-02, -1.0843e-02, ..., -2.8168e-02,
 -1.0555e-02, -7.3098e-02]]], grad_fn=<AdaptiveAvgPool2DBackward0>)
torch.Size([1, 512, 14, 14])
```

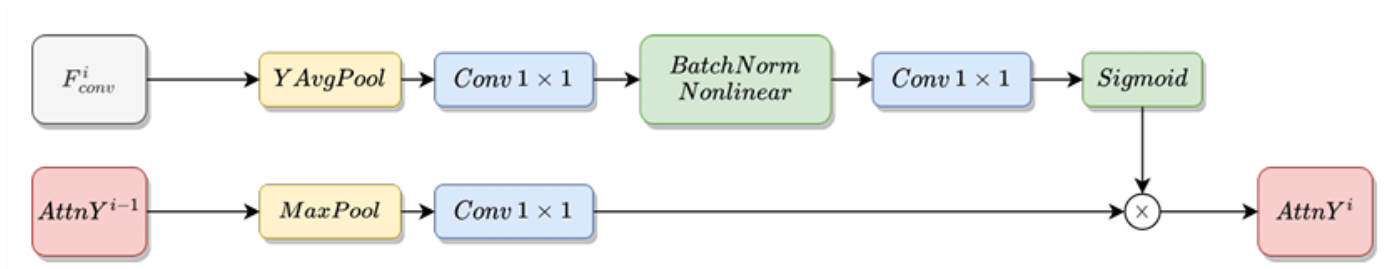
```
torch.Size([1, 512, 14, 14])
```



# 3. Progress After Interim

## Feature Extraction - CVA Module

- CVA: Continuous Vertical Attention
  - micro expression은 vertical한 특징에 크게 영향을 받음
  - Y Avg Pool을 통해 Vertical Feature를 강화한 attention map 제작
  - 이전 attention map과 각 layer를 거친 후의 data를 합쳐 다음 attention map을 제작
- 최종 attention map과 resnet의 결과를 element-wise multiplication



# 3. Progress After Interim

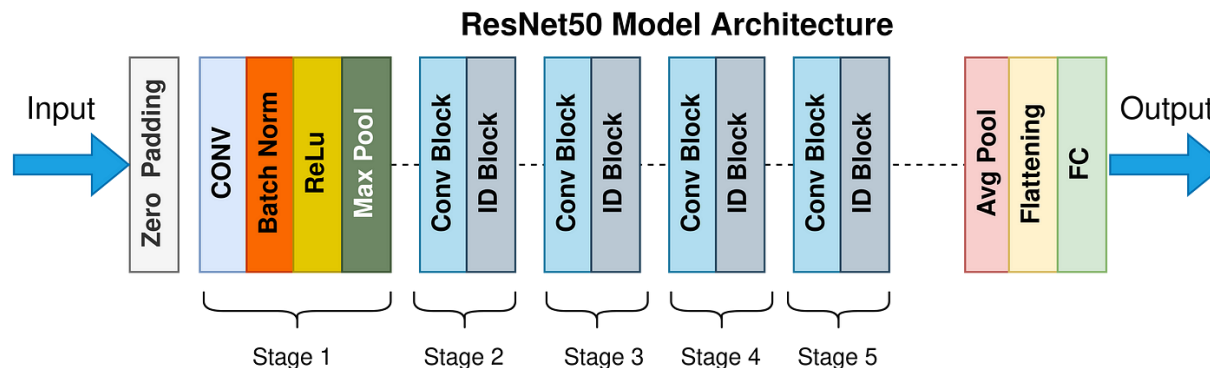
## Feature Extraction - FPF

- CNN Model: ResNet 50 Model
- Onset, Apex Frame을 각각 넣어 output 추출 → element-wise addition
- TMFE, FPF의 결과를 concat하여 classification part로 전송

Onset Frame



Apex Frame



```
tensor([[[[0.0000, 0.0000, 0.0000, ..., 0.0148, 0.1535, 0.0000],
[0.0000, 0.0000, 0.0000, ..., 0.6041, 0.7765, 0.3602],
[0.0000, 0.0000, 0.4996, ..., 1.6035, 2.1920, 1.6064],
...,
[1.6197, 1.8867, 1.9390, ..., 0.0000, 0.0000, 0.0000],
[1.6312, 2.1016, 1.0926, ..., 2.3249, 4.8814, 0.4198],
[0.0000, 0.0000, 1.9958, ..., 0.0000, 0.0000, 1.3007]]]])
```

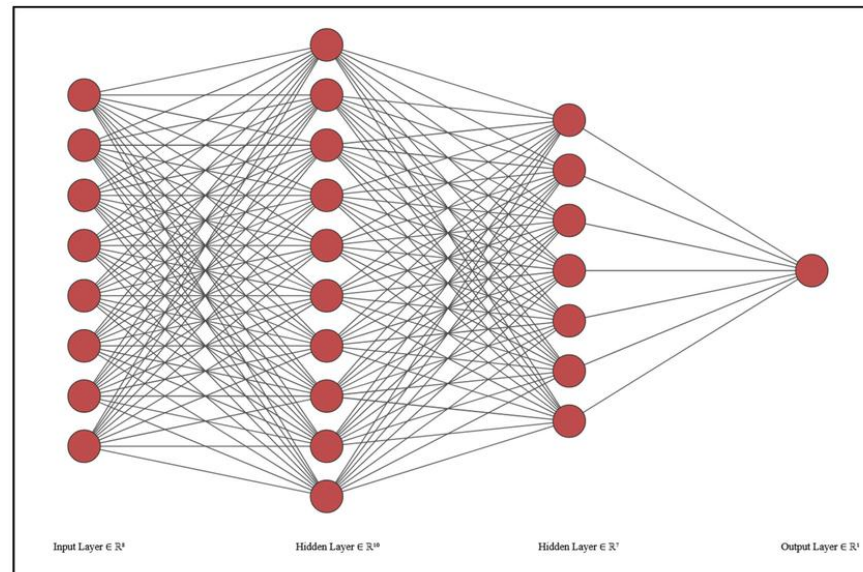
```
...
[[0.9767, 0.0000, 0.0000, ..., 0.0000, 0.0000, 0.0000],
[1.5177, 0.0000, 0.0000, ..., 0.0000, 0.0000, 0.0000],
[2.5181, 0.2819, 0.0000, ..., 0.0000, 0.0000, 0.0000],
...,
[1.0867, 0.0000, 0.3025, ..., 0.0000, 0.0000, 0.0000],
[3.0684, 1.1337, 0.6624, ..., 4.2551, 0.0000, 0.0000],
[0.0000, 2.2607, 0.0000, ..., 2.9232, 2.9928, 0.9128]]]],
grad_fn=<AddBackward0>)
torch.Size([1, 196, 14, 14])
torch.Size([1, 196, 14, 14])
```

# 3. Progress After Interim

## Binary Classification

- Binary Classification: implement binary classification model with logistic regression
- Input dim:  $708 \times 14 \times 14 + 29$  (concatenated features of FPF and TMFE + AU features)

```
Python Prompt (miniconda) x
C:\Users\woosu>python reset_18.py
tensor([[[[1.9122e-02, 0.0000e+00, 0.0000e+00, ..., 0.0000e+00,
0.0000e+00, 0.0000e+00],
[1.9366e-01, 0.0000e+00, 0.0000e+00, ..., 2.7309e-02,
6.6581e-02, 1.4099e-02],
[1.9006e-01, 2.9150e-02, 3.2769e-01, ..., 1.7621e-01,
2.6516e-01, 5.8970e-02],
...
[2.7978e-01, 2.2612e-01, 8.8220e-01, ..., 3.0217e-01,
3.2452e-01, 1.8696e-01],
[1.3331e+00, 2.5570e+00, 1.7875e+00, ..., 1.2830e+00,
3.4094e-01, 5.2197e-01],
[1.0831e+00, 1.2012e+00, 1.9679e+00, ..., 1.2875e-01,
1.5439e+00, 9.1817e-01]])])
..
[[[7.7813e-01, 1.0000e+00, 0.0770e-01, ..., 1.1621e+00,
1.2508e+00, 1.7970e+00],
[7.9256e-01, 2.0637e+00, 2.5812e+00, ..., 2.8303e-01,
1.9266e-01, 0.6860e-01],
[1.5204e+00, 2.2312e+00, 1.3143e+00, ..., 1.1831e+00,
1.0000e+00, 1.2003e+00],
...
[9.7813e-01, 1.0000e+00, 0.0000e+00, ..., 2.5077e-01,
3.7121e-01, 1.6077e+00],
[4.5127e-01, 0.0000e+00, 0.0000e+00, ..., 0.0000e+00,
0.0000e+00, 0.3150e-01],
[0.0000e+00, 0.0000e+00, 0.0000e+00, ..., 0.0000e+00,
0.0000e+00, 0.0000e+00]])])
torch.Size([1, 708, 14, 14])
```



```
C:\Users\woosu\miniconda3\
ill be removed in the futur
e using storages directly.
return self.fget.__get__
거짓말 가능성: 98.6114%
```

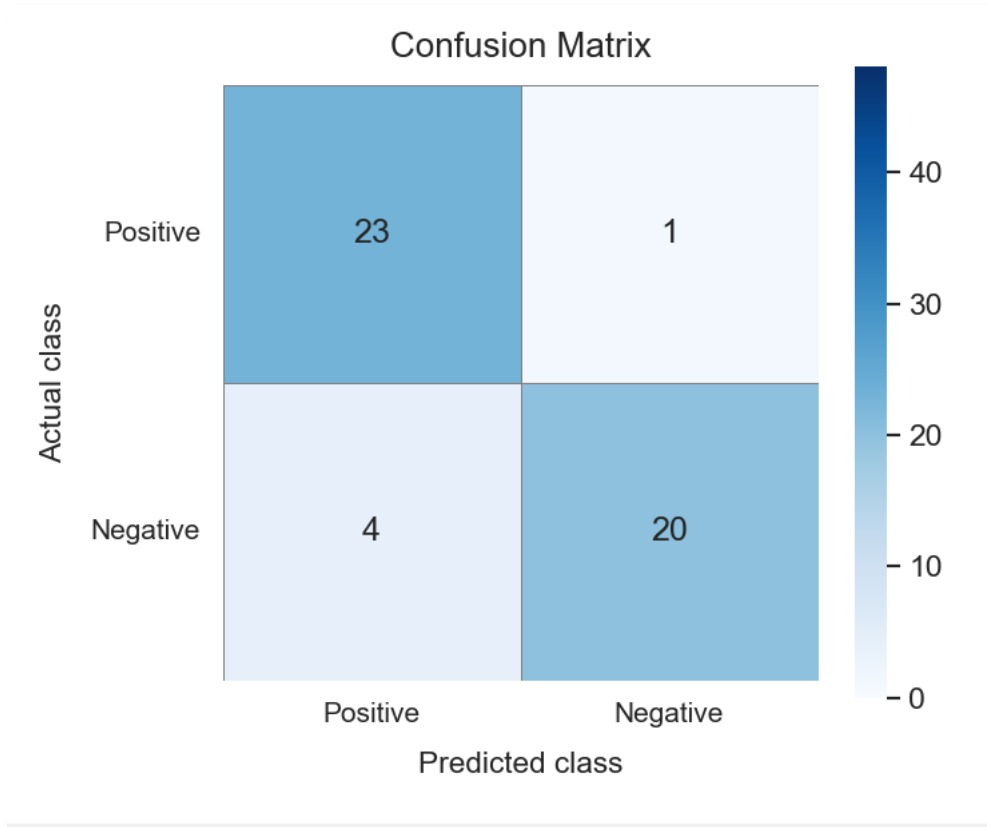
# **4. Evaluation**

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## 4. Evaluation

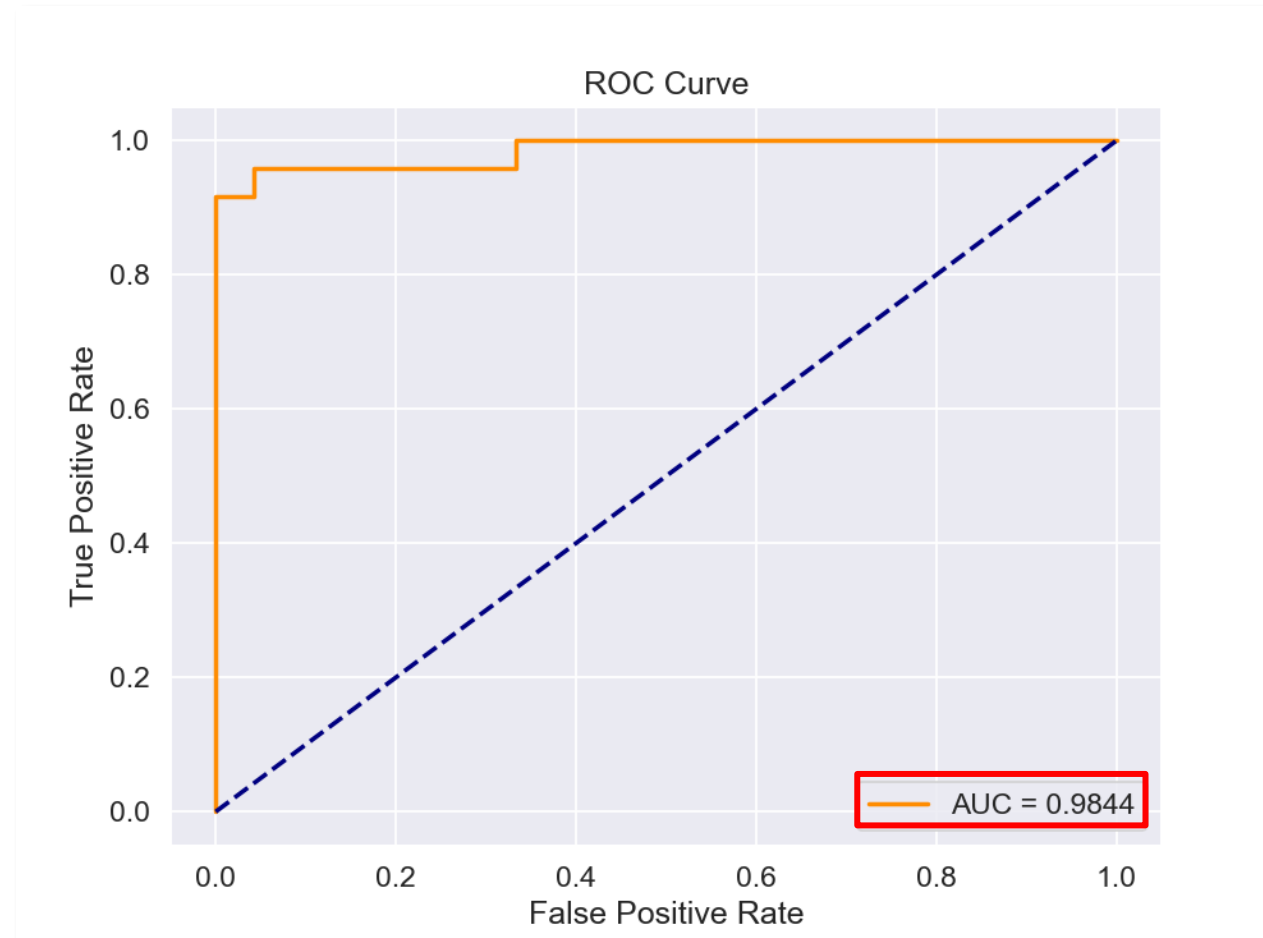
- Binary classification 성능 평가: confusion matrix와 ROC curve 활용
  - Actual class와 prediction 비교해 TP, TN, FP, FN 계산
    - confusion matrix 생성 및 accuracy, precision, recall, f1 계산
  - TPR, FPR 계산
    - ROC curve 생성 및 AUC score 계산
- 시각화: Confusion matrix, ROC curve 시각화

# 4. Evaluation



- TP: 23, FN: 1, FP: 4, TN: 20
- Accuracy = 0.8958
  - How many data were correctly predicted?
- Precision = 0.8519
  - How many data predicted as positive were correct?
- Recall = 0.9583
  - How many data of the actual positive class were correctly predicted?
- F1 score = 0.9020
  - How well did it predict the positive class in general?

# 4. Evaluation



# Thanks for listening

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ISOO

김준하, 노윤현, 윤지현, 조해나, 최우석

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# Reference

- <https://en.wikipedia.org/wiki/Microexpression>
  - <https://www.kaggle.com/search?q=FER+in%3Adatasets>
  - <https://www.kaggle.com/search?q=Micro-expression+in%3Adatasets>
  - <https://paperswithcode.com/task/facial-expression-recognition>
  - <https://paperswithcode.com/task/micro-expression-recognition>
  - <https://openreview.net/pdf?id=mHYkcQzdae>
  - [https://ai.google.dev/edge/mediapipe/solutions/vision/face\\_landmarker?hl=ko](https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker?hl=ko)
  - [https://github.com/ihp-lab/LibreFace/tree/main/AU\\_Detection](https://github.com/ihp-lab/LibreFace/tree/main/AU_Detection)
  - <https://arxiv.org/abs/2308.10713>
-